

**AI exposure
meets Spanish
labor markets**

AI Exposure and Labor- Market Outcomes in Spain

A research memo distilled into a 10–15 slide academic presentation

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The key question is not whether exposed jobs changed; it is whether AI caused the change

Spanish CNO4 occupations
matched to semantic AI exposure

Monthly SEPE panel
January 2021–March 2026

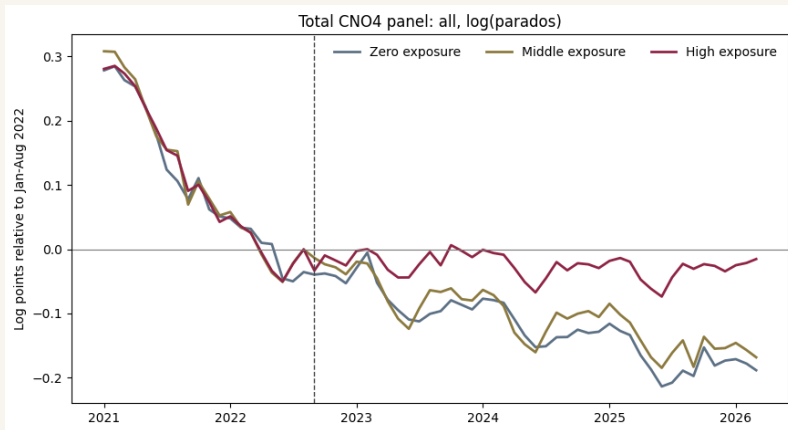
High AI exposure

Post-September 2022

Unemployment and contracts

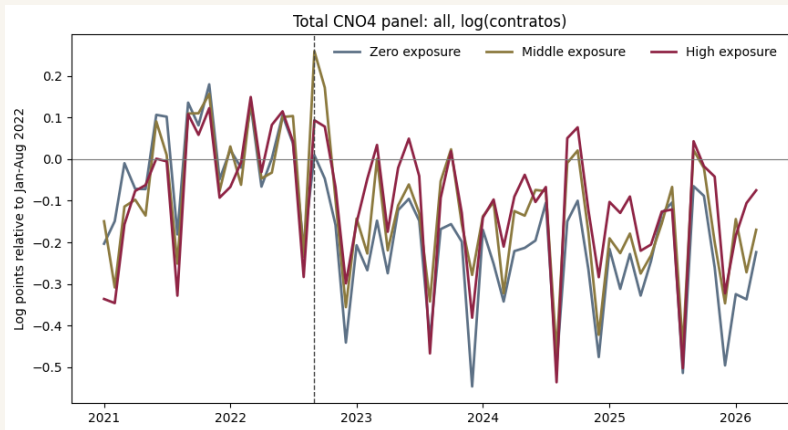
The empirical task is to separate a post-2022 occupational divergence from broader family-level shocks, churn, and reallocation.

High-exposure occupations diverge after 2022 in registered unemployment



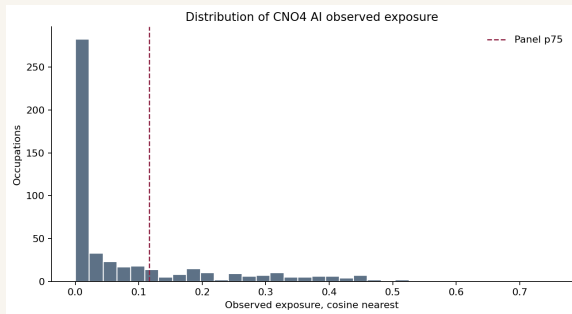
Indexed paths use the January–August 2022 mean as the normalization window. The figure is descriptive, not causal.

Contracts also rise, so the simple displacement story is incomplete



A rise in contracts in high-exposure occupations points toward churn, reallocation, or composition changes, not only lower labor demand.

The exposure measure is rich, but it is potential exposure rather than observed adoption



- Time-invariant CNO4 semantic exposure.
- Based on Spanish adaptation of Anthropic/O*NET exposure concepts.
- Best interpreted as task exposure, not monthly firm-level AI use.

Implication: the design tests whether exposed occupations moved differently after diffusion, not who adopted AI

The baseline TWFE estimate is positive and economically visible

0.089

log points in registered
unemployment

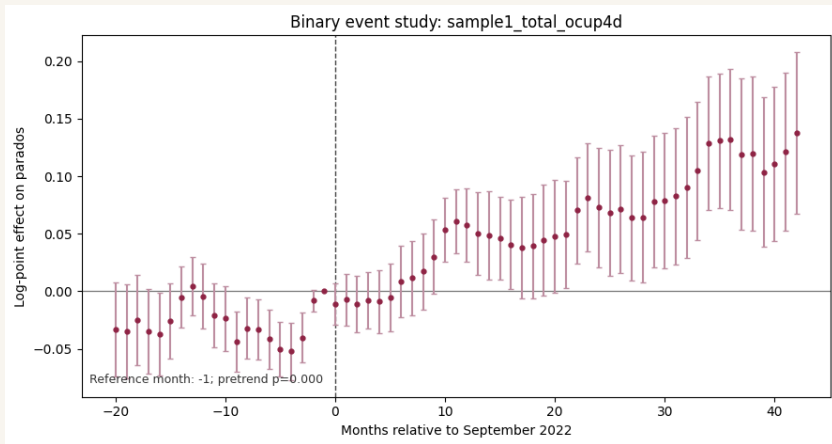
+9.3%

approximate post-2022 relative
increase

$$y_{ot} = \alpha_o + \lambda_t + \beta (HighAI_o \times Post_t) + \varepsilon_{ot}$$

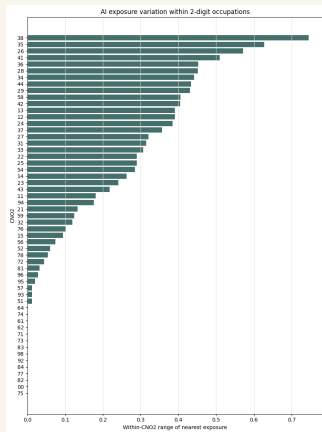
The coefficient compares high-exposure CNO4 occupations with zero-exposure occupations after September 2022, absorbing occupation and month fixed effects.

The baseline event study fails the parallel-trends test



This is the central reason not to present the baseline TWFE coefficient as causal.

Within-CNO2 comparisons are cleaner, but the identifying variation becomes thin

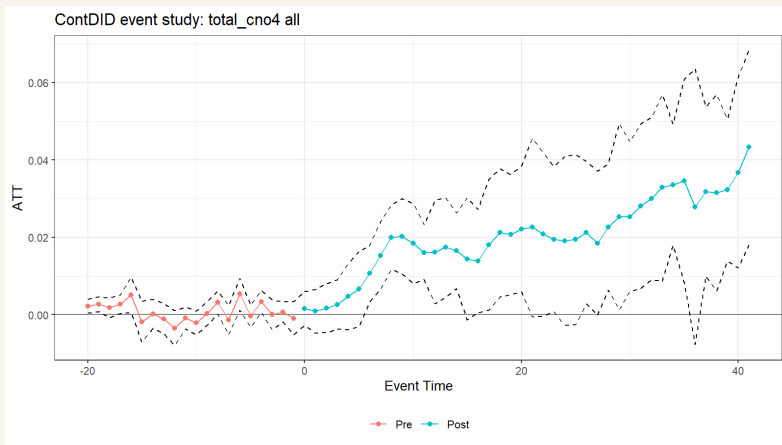


CNO2-by-month fixed effects compare detailed occupations within the same broad occupational family and month.

- This is conceptually attractive.
- It discards between-family variation.
- Power depends on within-family exposure support.

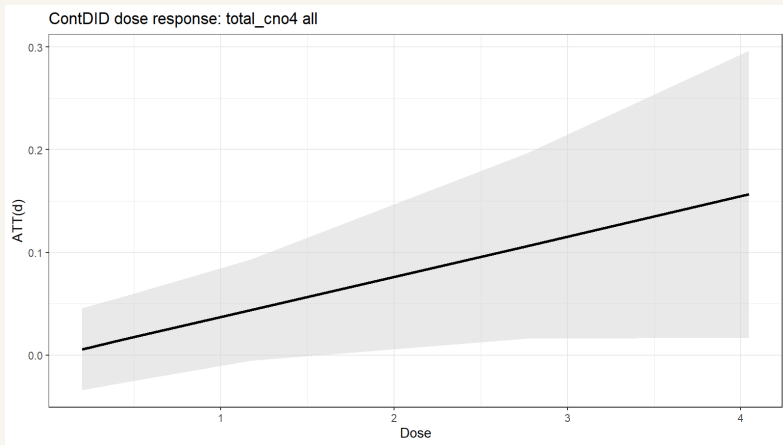
Interpretation: imprecision is not proof of zero effect; it shows the baseline depends heavily on broad occupational-family dynamics.

Continuous DiD uses the full exposure dose instead of a high-versus-zero cutoff



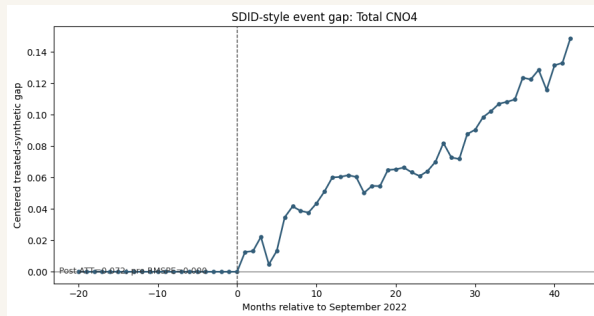
The contdid estimator is useful because it asks whether outcomes vary smoothly with the continuous exposure measure.

The dose-response pattern is suggestive, but support and functional-form checks matter



A dose-response plot is persuasive only where the exposure distribution has adequate support and pre-period balance is credible.

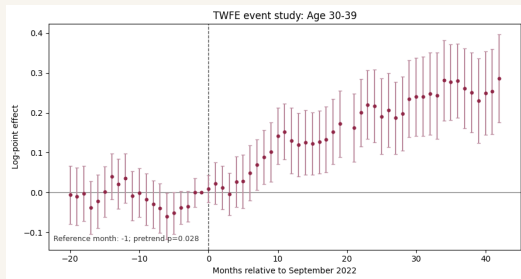
Synthetic DiD asks whether donors can reproduce the treated pre-period path



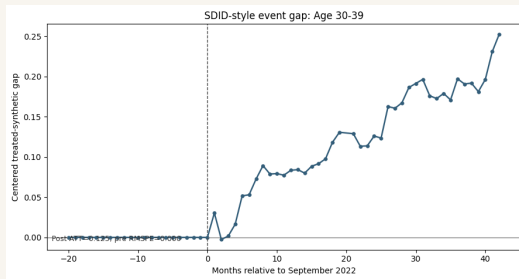
SDID is the right complement after TWFE pretrend failures because it forces the control group to look like the treated group before the diffusion window.

- Check pre-period fit.
- Inspect donor weights.
- Test sensitivity to donor pool restrictions.

Age heterogeneity points to the 30–39 group, but the result needs discipline



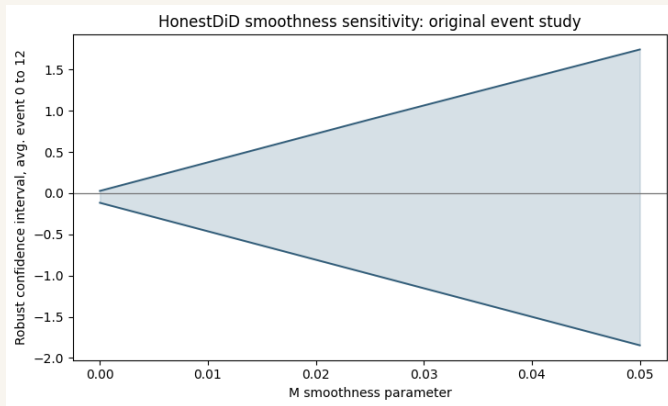
TWFE event study



SDID event study

Treat this as a priority robustness target, not as a standalone causal claim.

HonestDiD-style sensitivity should decide how much weight to put on the TWFE event study



Sensitivity analysis is especially important because the identifying assumptions are visibly strained in the baseline event study.

The most defensible conclusion is conservative

High-AI-exposure occupations show a positive post-2022 divergence, but the current evidence does not yet establish that AI exposure caused higher unemployment.

What is robust?

A descriptive post-2022 divergence in high-exposure occupations.

What is fragile?

Parallel trends and between-family occupational dynamics.

What comes next?

ContDID, SDID, support diagnostics, and donor-weight audits.

**The evidence is strongest
as a warning signal, not
yet as a causal verdict.**

The next version should make identification, not estimation, the center of the paper.